

SYED MUHAMMAD HAMZAH RIZVI

Forward Deployed Engineer

Manchester, UK | 07478 631940 | hamzahrizvi2@gmail.com | linkedin.com/in/hamzah-rizvi | github.com/hamzahrizvi | hamzahrizvi.github.io

PROFESSIONAL SUMMARY

Engineer with 3+ years owning customer deployments end to end across up to 15 concurrent projects spanning 10+ countries and thousands of operational endpoints — writing production code, going deep into core systems to unblock rollouts, and integrating complex software into unfamiliar architectures. Shipped a production RAG assistant now serving 300+ customer accounts weekly, backed by a 140+ test regression suite and a 34-case evaluation harness. Works shoulder-to-shoulder with customer engineering teams from scoping and building through to go-live, then acts as technical lead to engineering leadership. Hands-on with modern AI and AI-assisted development, driven by getting things shipped, and comfortable moving between deep technical work and explaining architecture to senior stakeholders.

CORE SKILLS

Engineering: Python, C++, FastAPI, REST API design and debugging, webhooks, authentication flows, JSON, distributed systems, production debugging across APIs, devices, firmware and networks, test-driven development, Git.

Applied AI: LLMs, agentic workflows, tool use, RAG, retrieval and reranking, grounding and hallucination reduction, model evaluation; daily AI-assisted development (ChatGPT, Claude, Copilot).

Deployment & Customer: End-to-end delivery, architecture discovery, integration and rollout, pilot ownership, technical lead to engineering leadership, technical documentation and enablement.

PROFESSIONAL EXPERIENCE

Technical Software Consultant — Innovative Technology, Manchester, UK

Mar 2023 – Present

- Own customer deployments of AI-powered biometric and identity-verification products end to end, from understanding each customer's architecture through integration, debugging and live production rollout.
- Pair with customer engineering teams to design and build REST API integrations, webhooks and authentication flows connecting AI products to their platforms and external systems.
- Go deep into the stack to unblock rollouts — diagnosing production issues spanning APIs, devices, mobile apps, firmware and networks — and feed fixes back into product and firmware.
- Act as technical lead to customer engineering and business stakeholders: scoping, solution reviews and rollout across international customers, balancing delivery, risk and timelines through go-live.
- Use AI coding assistants daily to prototype integrations, generate reference implementations and accelerate debugging, validating output against reproduced behaviour rather than accepting it blindly.
- Create reusable technical assets — integration guides, API documentation and diagrams — that make each deployment faster than the last.
- Delivered rollouts across retail, gaming and compliance-focused environments spanning thousands of operational endpoints.

Automation Engineer — Tripack Films Ltd., Pakistan

Mar 2020 – Sep 2021

- Built and maintained industrial automation workflows, integrating control systems and sensors; diagnosed hardware and software issues in real-time operational environments.

Software Consultant — Evolligence, Pakistan

Dec 2019 – Mar 2020

- Built and supported backend application logic using structured APIs and data-driven workflows, collaborating with development teams on delivery.

SELECTED ENGINEERING PROJECTS

GroundedOps — Production RAG Support Assistant — *Python, FastAPI, React, ChromaDB, Ollama, Docker, DeepSeek/OpenAI/Claude APIs*

- Deployed to production and now serving 300+ customer accounts weekly through a public embeddable widget and admin console; associated with a 40% reduction in first-line ticket volume and a 55% reduction in resolution time per internal Jira metrics since release (tracked alongside other concurrent initiatives, so not isolated to this tool alone).
- Built a retrieval-augmented QA system with hybrid retrieval (BM25 + dense embeddings via reciprocal rank fusion), sigmoid-calibrated cross-encoder reranking and NLI-based grounding checks to reduce hallucination from local LLMs.
- Diagnosed a query-routing bug by reproducing it in code, tracing it to an unscoped conversation-memory store and a weak follow-up heuristic; rebuilt the path with an LLM Rewrite-Retrieve-Read flow and per-session memory.
- Wrote a pure-stdlib regression suite of 140+ tests covering each diagnosed bug by name, plus an evaluation harness gating every change against 34 real test cases, enabling fast iteration without running model inference on every change.

Age Estimation & Demographic Analysis — *TensorFlow/Keras, EfficientNetB0, VGG16*

- Built a hierarchical multi-task EfficientNetB0 pipeline (gender, coarse age, fine age and continuous age regression) on 274k+ facial images from WIKI/IMDB, fine-tuned on UTKFace, with GPU-accelerated training and evaluation tooling; achieved 87.5% gender accuracy, 88.4% coarse-age accuracy, 92.6% near-group (± 1) accuracy and 8.52-year MAE.

EDUCATION

MSc Robotics and Automation (Minor in AI) — University of Salford, UK Graduated with Distinction. Dissertation: facial age and gender estimation using CNNs.	2021 – 2022
BEng Electrical Engineering — NUST, Pakistan Focus: computer vision, machine learning and robotics.	2015 – 2019